Unit 3, Lesson 7: Absolute Value

Benchmark

MA.6.NSO.1.3 Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers.

Learning Targets

- ☐ I can find the absolute value of a rational number.
- ☐ I can find the absolute value of a value and interpret it within the context.

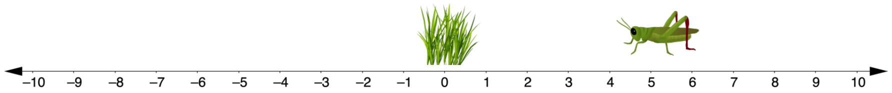
3.7.1 Warm-Up: Would You Rather?

1. Would you rather have Pile A or Pile B? Justify your reasoning.

Pile A	Pile B
14 pennies	2 pennies
7 nickels	4 nickels
2 dimes	2 dimes
1 quarter	2 quarters

3.7.2 Exploration: Jumping Bugs

A bug is jumping around on the number line. His home is located at 0. Use a cut out of the bug to answer the following questions about the bug's location.



1. The bug starts at 5 and jumps 3 units to the right.

   a. What is the bug's new location?

   b. How far away from home is the bug?

2.

   a. If the bug starts at 2 and jumps 5 units to the left, what is the bug's new location?

   b. How far away from home is the bug?

3. Mr. Rodriguez told his class to begin with the bug at home. Then the bug should jump 4 units. When Micah and Brianna did this, they discovered that their bugs were in two different locations. Discuss with your partner how this may have happened. Write your explanation or draw a picture of what could have happened.

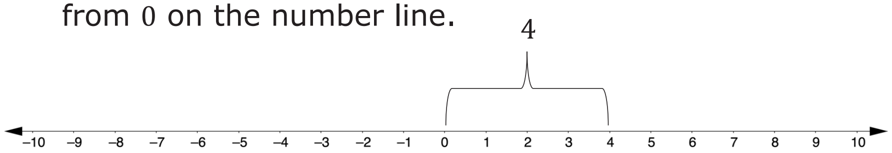
3.7.3 Guided Instruction: Absolute Value and Zero

Recall how opposite numbers relate to zero. Complete the explanation below by writing a word in each blank.

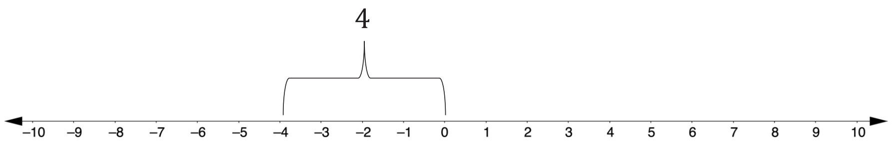
1. Pairs of opposite values are the same ________________ from zero on the number line. They are mirror images of each other, with a line of symmetry through ____________.

Absolute value is the distance a number is from zero (0) on a number line.

2. The absolute value of 4 is _______ because it is 4 units from 0 on the number line.



3. The absolute value of -4 is _______ because it is 4 units from 0 on the number line.



4. The values 4 and -4 are opposites. Each value is 4 units from zero. A number and its opposite have the same _____________________________.

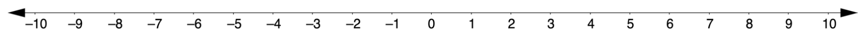
3.7.4 Guided Practice: Absolute Value and Zero

Absolute value is represented using the notation $|a|$. The notation $|a|$ is read, "the absolute value of a."

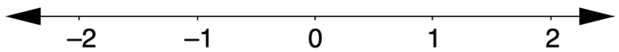
1.
   a. The $|-8| =$ _____ because the distance from -8 to 0 is _____.

   b. What other number has the same absolute value as -8?

   c. Use the number line to show the other number that has the same absolute value as -8.



2. On the number line, plot and label all numbers with an absolute value of $\frac{3}{2}$.



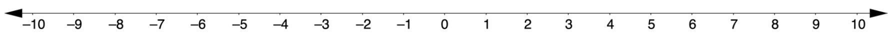
3. Hector and his partner, Salih, were given the following two problems to complete.

$|5.5| = \underline{\hspace{3em}}$ $\qquad\qquad$ $|\underline{\hspace{3em}}| = 3.2$

Hector and Salih had the same answer for one of the blanks but two different answers for the other. Their teacher told them they both had the correct answers.

a. Which problem can have more than one correct answer?

b. Use the number line to explain how this is possible.



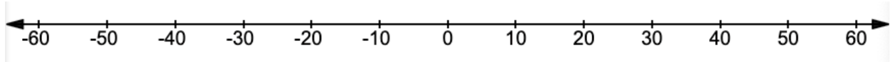
3.7.5 Collaboration: Absolute Value and Opposites in the Real World

1. A tiger was rescued by The Brevard Zoo in Melbourne, Florida. The zoo veterinarian examined the tiger and determined the animal’s weight to be 29.6 pounds below average.

   a. What value represents this situation?

   b. What is the absolute value of the value that represents the tiger’s weight loss?

   c. Model the tiger’s weight loss on the number line.



   d. What does 0 represent in this situation?

   e. If the tiger’s weight had been 29.6 pounds above average instead of below, what would be the same about this situation?

2. Simone is doing research on different rides at the Strawberry Festival. She attaches an accelerator to a bumper car to measure the velocity of her car. She recorded the velocities, in miles per hour (mph), in the table.

Speed is the absolute value of velocity.

Time (seconds)	Velocity (mph)	Time (seconds)	Velocity (mph)	Time (seconds)	Velocity (mph)
5	-3.1	20	-2.4	35	-5.2
10	-2.7	25	-6.3	40	3.1
15	4.8	30	2.7	45	-7.8

a. Determine when the bumper car is at the same speed. List the times and the speed of the car at those times.

b. At what time is the bumper car traveling at the slowest speed?

c. At what time is the bumper car traveling at the fastest speed?

3. Before Bryan goes snow skiing, he asks a friend about the weather. The friend says that the temperature is $5°\text{C}$ away from 0.

Select all the temperatures that are possible.

- ☐ $-10°\text{C}$
- ☐ $-5°\text{C}$
- ☐ $0°\text{C}$
- ☐ $5°\text{C}$
- ☐ $10°\text{C}$

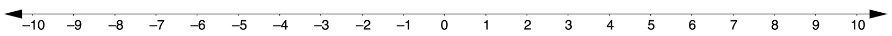
4. A part of the city of New Orleans is 19 feet below sea level. We can use "-19 feet" to describe its elevation, and "$|-19|$ feet" to describe its vertical distance from sea level. In the context of elevation, what do each of the following numbers describe?

Value	Meaning in Context
25 feet	
$\|25\|$ feet	
-8 feet	
$\|-8\|$ feet	

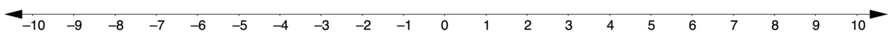
3.7.6 Your Turn!

1. Complete each statement and model it on the number line given.

   a. $|-7.8| =$ ________



   b. $|\_\_\_\_\_| = 3\frac{1}{2}$

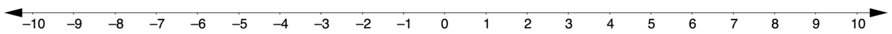


2. The absolute value of the elevation of Enterprise, Florida is 19.7 feet. In relation to sea level, write the two values that could represent the elevation of Enterprise, Florida.

3.7.7 Wrap-Up: Ticket Out the Door

1. Complete the statement and model it on the number line.

$$\left|-\frac{10}{4}\right| = \_\_\_\_\_\_\_\_$$



2. The number -220 represents a withdrawal in dollars from a checking account and 220 represents a deposit in dollars to the same checking account. What do the absolute values of these values mean in terms of this context?